OLIST E-COMMERCE

Revenue & Delivery Performance Analysis

Executive Summary

Period: September 2016 – October 2018

Orders Analysed: 99,441

Tools: PostgreSQL · Power BI

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PLATFORM SNAPSHOT

R$ 423,780	97.78%	R$ 139.93	71
Total Uncollected Revenue	Delivery Success Rate	Avg Order Value	Product Categories

THE BUSINESS PROBLEM

"We are losing revenue we should be collecting. Orders are getting canceled, going undelivered, or sitting in processing for days. We don't know how much money this represents or which categories and regions are worst affected. Our delivery estimates are also wrong. Late delivery is our number one driver of 1-star reviews. I need a clear picture: where is the money going, who is responsible, and what should we fix first."

— Olist CEO

THREE QUESTIONS THIS ANALYSIS ANSWERS

- **1.** How much revenue is being lost, what type, and which categories and regions are most affected?
- **2.** How frequently are delivery commitments being missed, by how much, and who is responsible?
- **3.** Where should intervention be prioritised to generate the highest return?

DATASET

- 99,441 orders cleaned from 99,992 raw records across 8 source tables
- 551 duplicate review records removed
- 1,362 impossible timestamps flagged and nulled
- 23 delivery-sequence violations flagged
- Period: September 2016 – October 2018

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REVENUE OVERVIEW

ACTUAL VS POSSIBLE REVENUE — MoM CHART

- Monthly collected revenue plotted against maximum possible revenue from January 2017 to October 2018
- The gap between the two lines represents monthly revenue leakage
- Outside seasonal peaks the gap remains proportionally stable — confirming the loss is **structural, not event-driven**

KEY FINDING — NOVEMBER 2017

- November 2017 recorded the **largest single monthly revenue gap** in the entire dataset
- **R$ 17,000 gap** between actual and possible revenue in one month alone
- Coincides with Brazil's Black Friday season — order volume spiked but operational capacity did not scale
- Result: concentrated cancellations, processing bottlenecks, and delivery failures in a single 30-day window
- **This is a recurring annual risk.** Without advance preparation the November 2017 pattern will repeat every peak season

REGIONAL PERFORMANCE — MAP

- States coloured by **revenue loss rate** — percentage of potential revenue not collected
- **Red** = highest loss · **Yellow** = moderate · **Green** = lowest loss

KEY FINDING — RORAIMA LEADS ALL STATES

- **Roraima (RR)** records a **10% revenue loss rate** — highest of any state in Brazil

States with highest loss rates:

- **RR — Roraima** → 10.0% | Highest in Brazil
- **AP — Amapa** → Elevated | Far north — logistics gap
- **AC — Acre** → Elevated | Far north — logistics gap
- **AM — Amazonas** → Elevated | Far north — logistics gap
- **AL — Alagoas** → Elevated | Northeast — courier failures
- **MA — Maranhao** → Elevated | Northeast — courier failures

- SP, RJ, and MG maintain the **lowest loss rates** despite the highest order volumes
- Delivery reliability and revenue performance are **directly and geographically correlated**
- The map answers the CEO's first question: **the money is going north — the cause is logistics infrastructure, not demand**

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CATEGORY AND PRODUCT ANALYSIS

LOSS REVENUE BREAKDOWN

- R$ 423,780 in uncollected revenue splits into three fundamentally different situations:

R$ 108,026 — Permanently Lost (25.5%)
Canceled and unavailable orders · Will never be recovered · Requires immediate intervention

R$ 177,370 — In Transit (41.9%)
Shipped but not yet delivered · Healthy pipeline · No action needed

R$ 138,382 — Stuck In Pipeline (32.6%)
Processing and invoiced states · Operationally blocked · Potentially recoverable

- **Only R$ 108,026 demands urgent intervention.** Conflating all three produces the wrong urgency and the wrong solution.

TOP 10 REVENUE LOSING CATEGORIES

The top 10 revenue losing categories have collectively cost Olist R$ 254,627 in lifetime revenue loss — representing 60% of all uncollected revenue on the platform.

Category	Revenue Lost	Share
watches gifts	**R$ 41,208**	16.2%
sports leisure	**R$ 38,399**	15.1%
health beauty	**R$ 29,158**	11.5%
cool stuff	**R$ 27,649**	10.9%
computers accessories	**R$ 26,548**	10.4%
furniture decor	**R$ 22,181**	8.7%
housewares	**R$ 20,005**	7.9%
garden tools	**R$ 17,073**	6.7%
bed bath table	**R$ 16,472**	6.5%
auto	**R$ 15,929**	6.3%

- **watches gifts** leads due to high average order value — moderate cancellation rate on expensive products produces large absolute losses
- **cool stuff** carries the highest cancellation RATE — impulsive orders canceled before dispatch · Requires friction-based solution
- watches gifts + cool stuff alone = **R$ 68,857 (27.1%)** of all permanently lost revenue from the top 10

MARKET OPPORTUNITY — SCATTER PLOT

- Identifies high price, low volume categories — the biggest **untapped revenue potential** on the platform

Small Home Appliances · Avg order value: R$ 660.44

Order volume too low despite highest avg price · Increase volume through promotion and homepage visibility · Highest revenue per order of all four categories

Furniture and Bedroom · Avg order value: R$ 226.45

Order volume below potential for high consideration purchase · Increase volume through seller recruitment and seasonal campaigns

Construction Safety · Avg order value: R$ 229.15

Avg price needs to increase alongside volume · Recruit premium sellers with higher quality ranges to lift transaction value

Construction Tools · Avg order value: R$ 174.00

Both avg price and volume below potential · Dual investment needed: seller quality improvement AND promotion · Longest term but largest addressable market

A 20% increase in order volume across these four categories could generate R$ 500,000+ in additional annual revenue — without acquiring a single new customer.

DELIVERY PERFORMANCE

FUNNEL ANALYSIS — ORDER JOURNEY

- **Purchased:** 99,441 orders → **100.00%**
- **Approved:** 99,280 orders → **99.84%**
- **Shipped:** 97,122 orders → **98.21%**
- **Delivered:** 96,007 orders → **97.02%**

- **3,434 orders never reached customers**
- Largest drop occurs between **shipped and delivered** — packages leave sellers correctly but fail in the last mile courier handoff
- First direct evidence: the delivery problem is in the **courier network**, not in seller preparation or order approval

LATE VS ON TIME ORDERS BY STATE

- Overall late delivery rate: **6.5%** (6,510 orders arrived after promised date)

States with highest late order rates:

- **AL — Alagoas** → Highest late rate on the platform
- **MA — Maranhao** → Second highest
- **AP — Amapa** → Elevated — far north
- **RO — Rondonia** → Elevated — far north
- **AC — Acre** → Elevated — far north
- **AM — Amazonas** → Elevated — far north

- **SP and RS** record the smallest late rates despite the highest order volumes — established courier routes perform significantly better than remote northern networks
- The late delivery map and revenue loss map show **identical geographic patterns** — same root cause: underdeveloped courier infrastructure in northern and northeastern Brazil

TOP 10 WORST SELLERS BY SHIPPING DELAY

- Delay range: **19 to 47.5 days** past contractual shipping handover deadline
- These are **not occasional lapses** — systematic repeated failures by a small number of sellers
- These 10 sellers contribute directly to the **20.2% of late deliveries** classified as seller fault
- Their customers wait **up to 47.5 days** beyond when packages should have left the warehouse
- **Immediate engagement** from seller management team required for each of these 10 sellers specifically

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FAULT TYPE AND CUSTOMER IMPACT

WHO IS CAUSING LATE DELIVERIES — FAULT TYPE DONUT

- Of **6,510 orders** that arrived after the promised delivery date:

Logistics Partner Fault · 5,191 orders · 79.7%
Courier failed in last mile · Seller shipped on time · Avg 10.9 days late

Seller Fault · 1,318 orders · 20.2%
Seller missed shipping handover deadline · Avg 9.7 days late

No Clear Fault · 1 order · 0.1%
Insufficient data to classify

THE CRITICAL IMPLICATION

- The courier network causes **4x more late deliveries** than sellers
- Penalising sellers fixes **20.2% of the problem at best** and risks alienating revenue-generating partners
- Renegotiating courier contracts addresses **79.7% of the problem** at its root cause
- The CEO's assumption that sellers are primarily responsible is **not supported by the data**

INTER-STATE VS INTRA-STATE DELIVERY COMPARISON

Metric	Inter-State	Intra-State
Total Orders	**60,922**	34,175
Late Rate	**8.1%**	4.6%
Avg Delay When Late	**11.5 days**	7.9 days

- Inter-state orders are **1.76x more likely to arrive late** than intra-state orders
- When they do fail they arrive **45% later on average** — 11.5 vs 7.9 days
- **61% of all platform orders are inter-state** — courier network failure affects the majority of Olist's transaction volume
- The solution is **contractual not operational** — renegotiate inter-state SLAs with binding penalty clauses

CUSTOMER RATING VS FAULT TYPE — SCATTER

The scatter proves the causal chain

- Early delivery (negative delay) → **avg review score: 4.2 stars**
- Late delivery (positive delay) → **avg review score: 2.8 stars**
- **1.4 star difference** driven entirely by delivery timing

- **Logistics fault orders dominate** the low score, high delay cluster — courier failures are the primary driver of 1-star reviews

- Seller fault orders appear in same cluster but proportionally smaller — consistent with their 20.2% share

THE CAUSAL CHAIN IS PROVEN BY THE DATA

- Courier failures → late delivery → low review scores → customer churn
- **Fix the courier network. Reviews improve automatically.**

RECOMMENDATIONS

PRIORITY ACTION PLAN

#	Action	Owner	Timeline	Impact
1	Renegotiate courier contracts Focus: AL, MA, RR, AM, AP, AC, RO	Operations Logistics	60-90 days	Reduce inter-state late rate from 8.1% to under 5% Improves reviews for 61% of all orders
2	Seller late fee structure Warning → fine → suspension within 90-day window	Seller Management	30 days	30-40% reduction in seller-caused delays within Q1
3	Cancellation fee for orders above **R$ 200** 5% of order value after 24-hour grace period	Product Team	2-3 weeks	Recover R$ 20,000-R$ 30,000 annually Targets watches gifts (R$ 41,208) and cool stuff (R$ 27,649)
4	Invest in opportunity categories **Small appliances · Furniture · Construction safety · Construction tools** Prioritise November window	Marketing Growth	45-60 days	R$ 500,000+ additional annual revenue at 20% volume uplift

DATA NOTES

- 99,441 orders cleaned from 99,992 raw records across 8 source tables
- 551 duplicate review rows identified and removed
- 1,362 impossible timestamps flagged and nulled in derived columns
- 23 delivery-before-carrier violations flagged
- 2018 data ends October — partial year, YoY comparisons adjusted accordingly
- Full cleaning documentation available in the project repository

GitHub: https://github.com/SarbajoyGhatak · LinkedIn: https://www.linkedin.com/in/sarbajoy-ghatak-717465272/